Risk Assessment

Dye Transfer in High-Temperature Container Storage

Background

Current Standard Test:

- **AATCC 163, Option 1**
- Simulated storage at **24 °C ± 3 °C for 48 hours**
- Purpose: Predictive test for temperature sensitivity under normal storage conditions.

Packaging Type:

- Garments packed with **paper inserts and paper bags**

Concern:

- Container temperatures during summer can reach **60~70 °C** in the worst-case high-temperature exposure
- Question: Does this pose a risk of **dye transfer** during transit?

Risk Factors

High Temperature (up to 70 °C):

Can accelerate dye migration, especially for dark or heavily dyed garments.

Moisture + Pressure:

Key drivers for dye transfer ; usually occur with **plastic packaging** or condensation.

Paper Packaging:

Breathable and absorbent -> reduces moisture accumulation and dye migration risk.

No solvent-like interaction (unlike plastic packaging)

Industry Feedback & Historical Data

Reported Issues:

Dye transfer complaints at 70 °C are **rare** when paper packaging is used.

Most cases occur with **polybag packaging combined with humidity.**

Internal Complaint Database:

No significant trend of color bleeding or staining linked to container heat exposure.

Conclusion

Risk Level: Low for current packaging configuration.

Reason: Paper packaging mitigates moisture and contact issues, even under high temperature.

In addition, our garment is normally with one solid color.

My POV:

No need to adopt 70 °C extreme test as a standard requirement.

Fabric applications may vary, it really depends on whether such fabric is going to be adopted under the heat source or not and at beginning stage, we should give a clear definition before adoption.

For example, if the fabric is intended for hot yoga, optional testing is required.

Optional test condition: To conduct DOE (Design Of Experiments) at 50/60/70 °C for 4 hours to confirm stability for certain critical products such as Nylon 66 made fabrics/garments with specific colors!